Human–AI Productivity Paradoxes

Aouad, Lykouris & Zhong (2026) — Proposition 2.1

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Repository 1 — August 2026

github.com/johnbarraza/ai-01-aouad

1. The paper

Question. Can AI reduce human effort while still increasing output?

Single mechanism. Skill, effort, and AI are perfect substitutes:

$$x = s + e + a.$$

Agent's problem.

$$\max_{e \geq 0} \; p(s + e + a) - \gamma e.$$

Economic intuition

Skill and AI each replace costly effort one for one. The constraint $e \geq 0$ creates the interior-versus-corner split.

2. Proposition 2.1 — result and conditions

Conditions: $s > 0$, $a \geq 0$, $\gamma > 0$; $p : \mathbb{R}_+ \to \mathbb{R}_+$ is weakly increasing, weakly concave, continuous, and twice differentiable; and

$$\limsup_{x \to \infty} \frac{p(x)}{x} < \gamma.$$

Define the critical input with the largest-maximizer tie-break:

$$x^* = \max \arg\max_{x \geq 0} \{p(x) - \gamma x\}.$$

Then

$$e^*(s,a) = (x^* - s - a)_+, \qquad p^*(s,a) = \max\{p(x^*), p(s+a)\}.$$

One-sentence intuition

The worker tops total input up to x^*; once $s + a$ reaches that target, effort falls to zero.

3. What I did

1. Rewrote the choice using total input:

$$\max_{x \geq s+a} \{p(x) - \gamma x\} + \gamma(s + a).$$

2. Verified the two feasible cases instead of assuming an interior FOC:

$$x^{\mathrm{opt}} = \begin{cases} x^*, & s + a \leq x^*, \\ s + a, & s + a > x^*. \end{cases}$$

3. Numerically audited the formula for $p(x) = 10(1 - e^{-0.4x})$ in `verify_proposition.py`.
4. Checked possible extensions: convex effort cost is already in Appendix D; a two-period non-myopic agent remains a tractable candidate.

4. Where I did not believe the AI

AI claim. The first-order condition $p'(s+e+a)=\gamma$ proves the result.

Verdict: incomplete.

- It starts by assuming an interior optimum.
- It misses the corner $e=0$.
- It does not handle a flat maximizing set or justify the paper's largest-maximizer tie-break.

My check. I derived the feasible set $x \geq s+a$ and verified both cases directly from concavity.